

SMART WEDDING PLANNING SYSTEM

Higher National Diploma in Software Engineering

Final Report Dissertation
HNDSE25.1F



School of Computing
National Institute of Business Management
Colombo - 07

**NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
HIGHER NATIONAL DIPLOMA IN SOFTWARE ENGINEERING**

Final Report Dissertation

Smart Wedding Planning System

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Date : Friday, March 27, 2026

The project is submitted in partial fulfillment of the requirement of the Higher National Diploma in Software Engineering HNDSE25.1F of National Institute of Business Management.

DECLARATION

We hereby declare that this dissertation represents our own work, conducted under the supervision of **Mr. Sampath Gamage**, and has not been previously included in a thesis or dissertation submitted to this or any other institution for a degree, diploma, or other qualifications. We have adhered to the ethical guidelines of **National Institute of Business Management (NIBM – Sri Lanka)** and acknowledge all contributions made towards the completion of this research.

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ABSTRACT

Wedding planning is a complex and time-consuming process that involves coordinating multiple services such as venues, catering, photography, decorations, and entertainment. Traditional wedding planning methods often rely on manual communication, fragmented information sources, and inefficient coordination between stakeholders. These limitations frequently result in delays, increased costs, and unsatisfactory outcomes for customers.

To address these challenges, the MarryMate Wedding Planning Platform was developed as a web-based solution that provides a centralized system for managing all aspects of wedding planning. The system integrates key features such as vendor discovery, budget management, booking services, and communication between users and vendors. Customers can easily browse available vendors, filter them based on preferences, and manage their wedding plans effectively.

A unique feature of the system is the integration of web scraping techniques, which automatically collect vendor information from external sources. This ensures that the platform provides updated and diverse vendor options without relying solely on manual data entry.

The system is developed using modern web technologies, including HTML, CSS, and JavaScript for the frontend, and Firebase services such as Authentication, Firestore Database, and Storage for backend operations. The final product is a scalable, user-friendly, and efficient platform that enhances the overall wedding planning experience.

ACKNOWLEDGMENTS

We would like to express our sincere gratitude to our supervisor, **Mr. Sampath Gamage**, for his invaluable guidance, encouragement, and continuous support throughout the development of this project. His insights and constructive feedback greatly contributed to the successful completion of this study.

We also extend our appreciation to the **National Institute of Business Management** for providing the academic environment and resources necessary to carry out this project.

Finally, we would like to thank our families, friends, and colleagues for their encouragement, patience, and support during the entire project period.

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Chapter 1: Introduction

1.1 Introduction to the Domain

The wedding industry represents a significant social and economic sector, involving multiple services such as venues, catering, photography, decoration, entertainment, and event coordination. Planning a wedding requires managing numerous vendors, schedules, financial decisions, and logistical arrangements. Traditionally, this process relied heavily on manual coordination, word-of-mouth recommendations, and physical visits to service providers.

In Sri Lanka, wedding planning remains largely fragmented, requiring couples to contact multiple vendors separately, compare services manually, and manage bookings through phone calls and in-person meetings. This approach is time-consuming and often leads to confusion, scheduling conflicts, and budget overruns.

With the rapid growth of digital platforms and online service marketplaces, there is an increasing demand for centralized solutions that simplify event planning. A digital wedding planning platform can streamline vendor discovery, scheduling, budget tracking, and communication, improve efficiency and reducing stress for couples.

MarryMate is designed to address these needs by providing a centralized web-based platform that connects couples with verified wedding service providers and enables efficient planning and coordination.

1.2 Research Gap

Despite the availability of event management platforms, several limitations still exist. Most systems do not provide intelligent budget planning features, making it difficult for users to manage expenses effectively. Additionally, many platforms lack proper integration between different services, forcing users to rely on multiple sources.

Another significant gap is the lack of automated data collection. Most platforms depend on manually entered vendor data, which can be incomplete or outdated. This reduces the reliability and usefulness of the system.

The MarryMate platform addresses these gaps by introducing automated data collection, centralized services, and intelligent planning tools.

1.3 Problem Definition

Traditional wedding planning involves several challenges, including difficulty in finding reliable vendors, poor budget management, and inefficient communication between stakeholders. Customers often spend a significant amount of time searching for services and comparing options.

Vendors face difficulties in reaching potential customers and managing bookings efficiently. Administrators also struggle to maintain accurate vendor information and ensure service quality.

This project aims to solve these problems by providing a digital platform that streamlines the entire wedding planning process.

- Fragmentation: No single platform exists to manage all vendors, appointments, budgets, and communications, forcing couples to use multiple disconnected tools.
- Time Inefficiency: Significant time is wasted on travel between vendors, repeated communications, and manual schedule coordination.
- Financial Inefficiency: Hidden costs, travel expenses, and lack of price transparency led to budget overruns.
- Information Asymmetry: Difficulty in accessing genuine reviews, verified service quality, and competitive market pricing.
- Coordination Complexity: Managing multiple vendors, payment schedules, and appointment timelines creates unnecessary stress and potential conflicts.
- Geographical Limitations: Couples in suburban areas have limited access to quality service providers and often must travel to urban centers.

1.4 Research Objectives

The main objective of this project is to develop a web-based wedding planning platform that simplifies and enhances the planning process. The specific objectives include:

- To create a centralized system for vendor discovery
- To implement a smart budget management system
- To enable effective communication between users and vendors
- To automate vendor data collection using web scraping
- To develop a scalable and user-friendly system

1.5 Proposed Solution

The proposed solution is a comprehensive web application that integrates all wedding planning functionalities into a single platform. The system provides separate dashboards for customers, vendors, and administrators.

Customers can search for vendors, manage budgets, send booking requests, and communicate with service providers. Vendors can manage their profiles and services, while administrators oversee system operations and vendor approvals.

1.6 Chapter Summary

MarryMate represents a significant advancement in digital solutions for the wedding industry, directly addressing the core inefficiencies that plague traditional planning methods. By providing a centralized, transparent, and efficient digital ecosystem, it empowers couples to plan their special day with reduced stress, better budget control, and increased confidence in their vendor selections.

This project not only demonstrates our technical capabilities as software engineering students but also highlights our ability to identify genuine market needs and develop practical, user-centered software solutions that have real-world impact.

Chapter 2: Methodology

2.1 Data Collection Method

To ensure that MarryMate addresses real-world wedding planning challenges, several data collection methods were used to understand user needs and industry practices.

- **Interviews**

Informal discussions were conducted with recently married couples and event planners to understand the difficulties faced during wedding preparation, vendor selection, and scheduling.

- **Online Research**

Existing wedding planning platforms and vendor websites were studied to identify common features, service presentation methods, and pricing models.

- **Surveys**

Feedback was collected from potential users to understand expectations, preferred features, and common planning challenges.

- **Market Observation**

Local wedding service providers' operations and customer interactions were observed to understand traditional workflows and service coordination practices.

These methods provided valuable insights that guided system requirements and feature design.

2.2 Software Process Model

The **Agile Software Development Model** was selected for developing MarryMate due to its flexibility and iterative approach.

Agile allows development to be completed in small increments, enabling continuous improvement and adaptation based on feedback.

Reasons for Choosing Agile

- Allows flexibility in refining features during development
- Supports continuous feedback and improvement

- Helps identify and fix issues early
- Encourages teamwork and effective collaboration
- Enables incremental delivery of system components

The project was developed in iterative phases, including planning, design, development, testing, and refinement.

2.3 Software Development Tools

The following tools and technologies were used to design and develop the MarryMate system.

❖ Development Environment

Visual Studio Code - Used as the primary code editor for frontend and backend development.

❖ Frontend Technologies

HTML - structure of web pages

CSS – styling and layout

JavaScript – interactive user interface functionality

❖ Database Management

Firestore - NoSQL document database

❖ UI/UX Design

Figma - Used to design user interfaces and create layout prototypes before development.

2.4 Testing Strategies

Testing was conducted to ensure the system functions correctly, securely, and efficiently.

- **Unit Testing:**
Individual components such as login validation and booking functions were tested independently.
- **Integration Testing:**
Ensured proper interaction between the frontend interface, backend logic, and database.
- **System Testing:**
The complete system was tested to verify that all features function together correctly.

- **User Acceptance Testing (UAT):**
Potential users tested the system to evaluate usability, navigation, and overall experience.
- **Security Testing:**
Authentication and user data protection mechanisms were tested to ensure secure access.
- **Cross-Browser Testing:**
The system was tested on multiple web browsers to ensure compatibility and consistent performance.

2.5 Implementation Plan

A phased implementation strategy was used to ensure systematic development and integration of system components.

Phase 1: Requirement Analysis & Planning

Requirement analysis is the initial phase of the project where the needs and expectations of users are identified and documented. In this stage, information is gathered through research, observations, or discussions to understand what the system should do. For this project, requirements included features such as vendor management, booking functionality, budget planning, and user communication. This phase ensures that the system is built according to user needs.

Phase 2: System Design

System design involves planning how the system will be structured and how different components will interact with each other. This includes designing the system architecture, database structure, and user interfaces. In this project, design activities included creating system architecture diagrams, defining database collections (such as users, vendors, bookings), and designing user-friendly interfaces. This phase acts as a blueprint for development.

Phase 3: Development

Development is the phase where the actual system is built based on the design specifications. It involves writing code, creating user interfaces, and integrating backend services. In this project, development was carried out using HTML, CSS, and JavaScript for the frontend, while Firebase was used for backend functionalities such as authentication and database management. All system features were implemented during this stage.

Phase 4: Testing & Debugging

Testing is conducted to ensure that the system works correctly and meets the required standards. It involves identifying and fixing errors or bugs in the system. Different testing methods such as unit testing, integration testing, and user interface testing were used in this project to verify functionality, performance, and usability. This phase ensures the reliability and quality of the system.

Phase 5: Deployment & Evaluation

Deployment is the final stage where the completed system is made available for users. The system is hosted on a platform so that it can be accessed through the internet. In this project, deployment was done using Firebase Hosting, allowing users to access the wedding planning platform online. This phase ensures that the system is ready for real-world use.

2.6 Functional and Non-Functional Requirements

2.6.1 Functional Requirements

1. AI Automation & Web Scraping (Core Innovation)

- The system allows vendors to input their website URL to automatically extract their business details (Company Name, Contact Person, Email, Phone, Description, and Profile Image) using an AI-powered scraping API.
- The system shall automatically crawl vendor websites to identify, extract, and categorize their service packages (Title, Price, and Images) and store them in the database.
- The system shall execute a scheduled background task (Cron Job) daily at 2:00 AM to synchronize vendor prices.
- The automated updater shall perform a "Safe Merge Synchronization," updating only the prices of existing packages without overwriting manual modifications made by the vendor.

2. Customer Features & Smart Planning

- The system shall feature a "Smart Budget Planner" that accepts a total budget and automatically allocates funds across standard wedding categories using predefined percentage logic.
- The Smart Budget Planner shall automatically recommend specific, active vendors that match the user's allocated budget and preferred district.
- The system shall allow customers to search, filter, and browse vendors by category (e.g., Photography, Venues) and district.

- The system shall allow customers to save (favorite) vendors, send direct messages, and submit booking/payment requests for specific vendor packages.
- The system shall calculate and display a dynamic "Wedding Timeline" countdown based on the customer's saved wedding date.

3. Vendor Management

- The system shall support two distinct registration flows: "Company Vendor" and "Individual Vendor/Freelancer".
- The system shall provide vendors with a dashboard to manually add, edit, or delete their services/products, by passing the AI scraper if desired.
- The system shall allow vendors to view, approve, or reject incoming customer booking requests and track payment statuses.

4. Authentication & Security

- The system allows users (Customers and Vendors) to register and log in securely using Email/Password or Google Single Sign-On (OAuth).
- The system shall enforce password complexity rules during registration (minimum 8 characters, uppercase, lowercase, number, and special character).

2.6.2 Non-Functional Requirements

These define the quality, performance, and security constraints of the platform.

1. Performance & Speed

- The AI Web Scraping API must process a vendor's website and return the extracted JSON data to the frontend UI within 10 seconds.
- Dashboard statistics, charts, and vendor grids must be loaded and rendered within 2 seconds for a seamless user experience.

2. Security & Privacy

- The backend Cron Job microservice must authenticate with the Firebase Database securely using a hidden Firebase Admin SDK private key, ensuring no unauthorized database access.
- The system shall implement a no-referrer policy to securely bypass third-party image hotlink protections (such as Cloudflare Error 1011) without exposing origin tracking data.

- API keys (such as the Google Gemini API key) must be stored securely in server-side environment variables (.env) and excluded from version control systems (Git).

3. Reliability & Availability

- The backend scraping service shall implement a robust retry mechanism (exponential backoff) with randomized User-Agent headers to handle network timeouts and prevent IP blocking by target websites.
- The system infrastructure (hosted via Firebase) shall aim for 99.9% up time to ensure vendors and customers can access their wedding plans at any time.

4. Usability & User Experience (UX)

- The user interface shall be fully responsive (Mobile, Tablet, Desktop) utilizing a modern "Glassmorphism" design system.
- The system must provide real-time UI feedback, including loading spinners during AI extraction and color-coded toast notifications for success/error states.

5. Scalability

- The NoSQL database architecture (Firestore) must be structured to support concurrent read/write operations for thousands of active users without performance degradation.

2.7 Chapter Summary

This chapter presented the methodologies and approaches used in the development of the MarryMate Wedding Planning Platform. It explained how data was collected using both manual research and automated techniques such as web scraping to ensure accurate and up-to-date vendor information.

The chapter also described the use of the Agile software development model, which allowed the project to be developed in iterative stages with continuous improvements based on feedback. Additionally, it outlined the technologies and tools used in the development process, including HTML, CSS, JavaScript, and Firebase services for backend operations.

Furthermore, the chapter discussed the testing strategies applied to ensure system reliability, including unit testing, integration testing, and user interface testing. The implementation plan was also highlighted, showing the structured approach followed from requirement analysis to deployment.

Overall, this chapter demonstrated how proper planning, appropriate tools, and systematic methodologies contributed to the successful development of a reliable, efficient, and user-friendly system.

Chapter 3: Analysis

3.1 Problem and Solution Analysis

The system was carefully designed to address the key problems identified during the analysis phase by introducing a centralized and automated solution. Instead of relying on manual and disconnected processes, the platform integrates all functionalities into a single system, allowing users to access services, manage information, and communicate efficiently in one place. This centralization reduces complexity, saves time, and improves overall coordination between customers, vendors, and administrators.

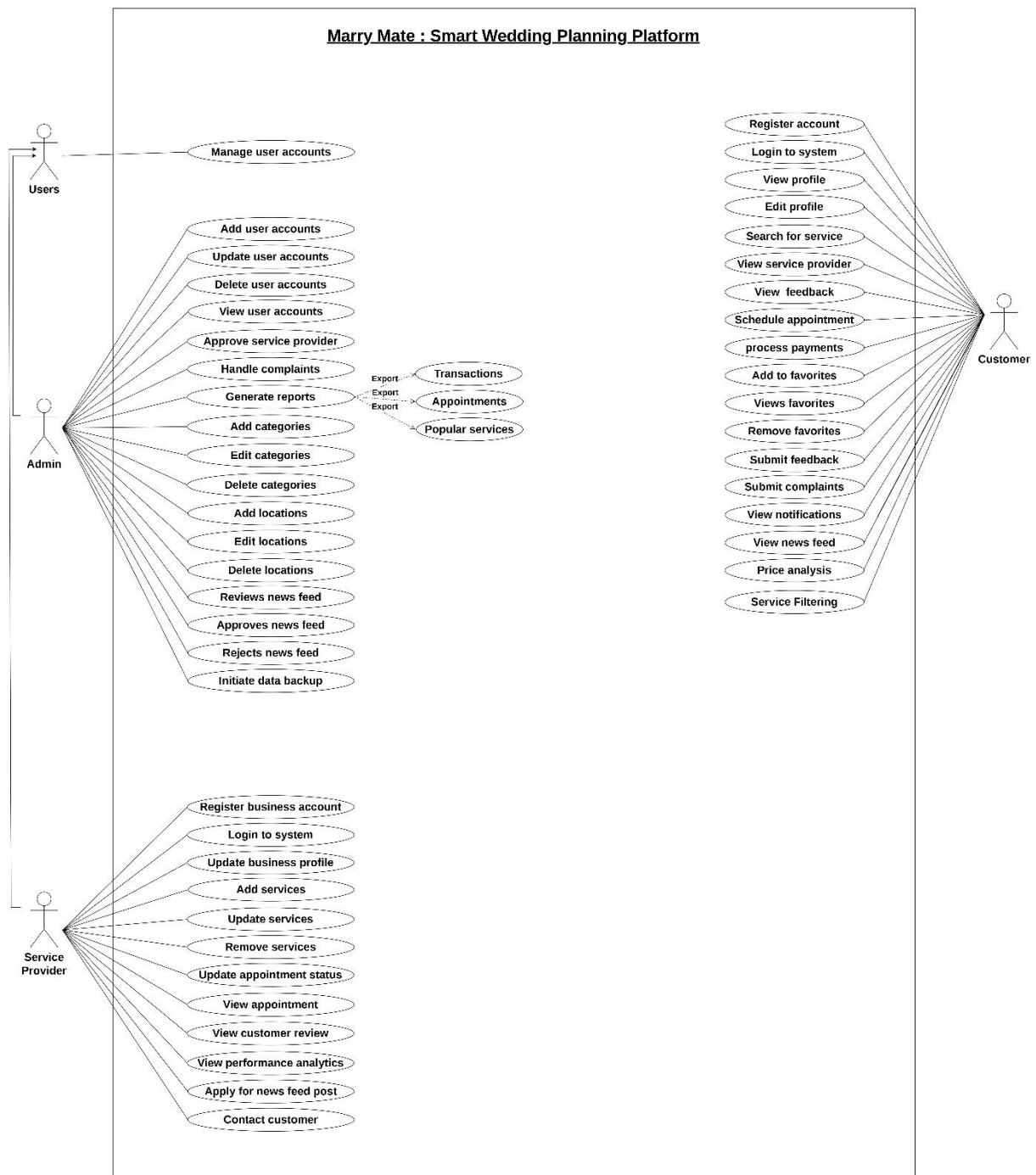
Furthermore, each identified problem was systematically analyzed and linked to a specific system feature that provides a practical solution. For example, the difficulty in finding reliable vendors is addressed through a vendor discovery module, while poor budget management is solved using a smart budget planner. Similarly, communication issues are resolved through an integrated messaging system. This problem-to-solution mapping ensures that every feature of the system directly contributes to solving real-world challenges, making the platform effective, efficient, and user-focused.

3.2 UML Diagram

Unified Modeling Language (UML) diagrams were used to visually represent the structure and functionality of the MarryMate system. These diagrams help illustrate how users interact with the system and how different components work together.

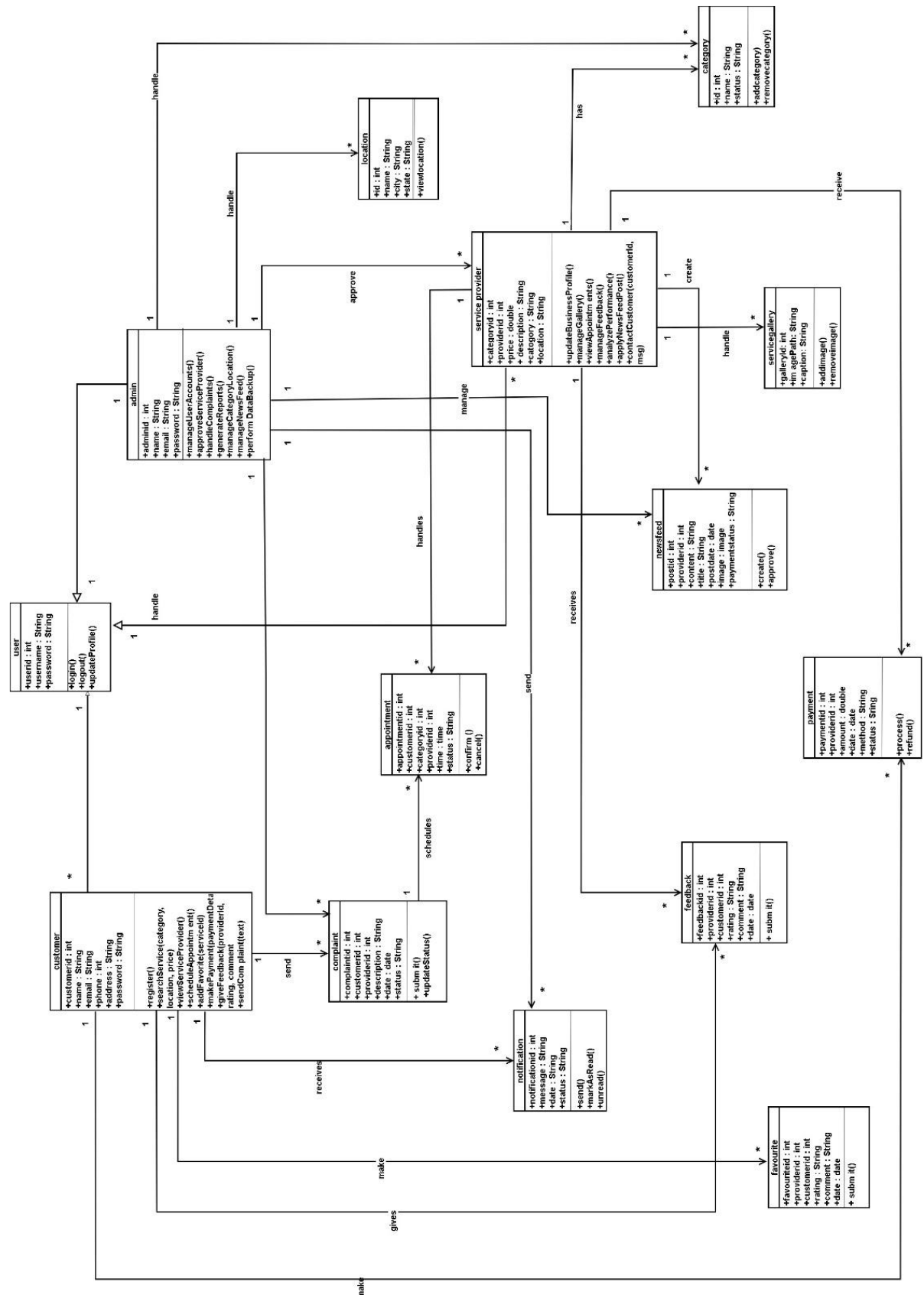
3.2.1 Use Case Diagram

The Use Case Diagram represents interactions between system users and the platform.



3.2.2 Class Diagram

The Class Diagram illustrates the system's object structure and relationships between entities.



3.3 Data Management of the Proposed System

The system uses **Firebase Firestore** to manage and store data efficiently. Firestore provides a NoSQL database that allows structured storage in collections and documents, ensuring scalability and real-time data access.

Key Data Collections:

- **Users:** Stores customer and vendor profiles, login credentials, and preferences.
- **Vendors:** Includes vendor services, ratings, pricing, and availability.
- **Bookings:** Tracks scheduled events, confirmations, and statuses.
- **Messages:** Stores communication between customers, vendors, and admin.
- **Budget Data:** Tracks expenses, budget limits, and automated suggestions

3.4 Chapter Summary

This chapter examined the challenges associated with traditional wedding planning, including difficulties in vendor discovery, inefficient scheduling, lack of transparency, and poor coordination between customers and service providers. These issues highlight the need for a centralized digital solution to simplify and streamline the planning process.

The proposed MarryMate system was analyzed as a solution that integrates vendor search, booking management, budgeting, and communication within a single platform. UML diagrams were introduced to illustrate user interactions, system structure, and relationships between system components, providing a clear understanding of how the platform operates.

In addition, data management strategies were discussed to ensure data integrity, security, and controlled access to system information. This analysis establishes the technical foundation necessary for the design and development of the system described in the next chapter.

Chapter 4: Solution Design and Development

4.1 System Architecture

The system follows **client-server architecture**, with the frontend handling user interaction and the backend managing data storage, authentication, and server logic. Firebase services integrate seamlessly with the frontend to provide real-time updates and secure operations.

Architecture Highlights:

- **Frontend:** HTML, CSS, and JavaScript for responsive design and dynamic features.
- **Backend:** Firebase Authentication, Firestore Database, and Storage for media and files.
- **Client Requests:** Customers and vendors interact with the system via dashboards, forms, and messaging.
- **Server Response:** Backend validates requests, updates the database, and provides feedback in real-time

4.2 Interface Design

The user interface was designed to be simple, intuitive, and user-friendly to ensure ease of navigation for all users.

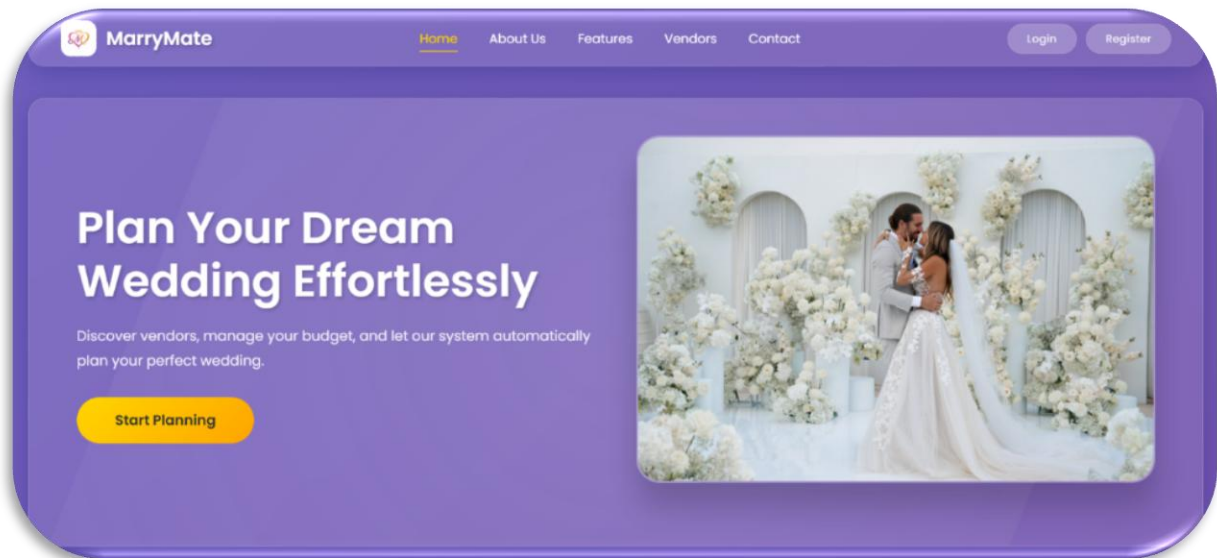
❖ Home Page

Purpose:

Provides an overview of MarryMate and allows users to explore available wedding services.

Key Features:

- Navigation menu
- Featured services
- Login & registration options



❖ User Registration Page

Purpose:

Allows new users to create accounts.

Key Features:

- Input fields for personal details
- Secure password creation
- Validation for required fields

The screenshot shows the 'Create Account' page on the MarryMate website. The page has a purple gradient background. At the top, there is a MarryMate logo and the heading 'Create Account'. Below the heading is a subtext 'Join MarryMate and start planning your dream wedding'. The form contains several input fields: 'Full Name', 'Email Address', 'Phone Number', 'Address', 'mm/dd/yyyy' (for date), and 'Password'. Each field has a corresponding icon (person, envelope, phone, location pin, calendar, and lock) and a placeholder text. There is also a 'Show/Hide Password' toggle button next to the password field.

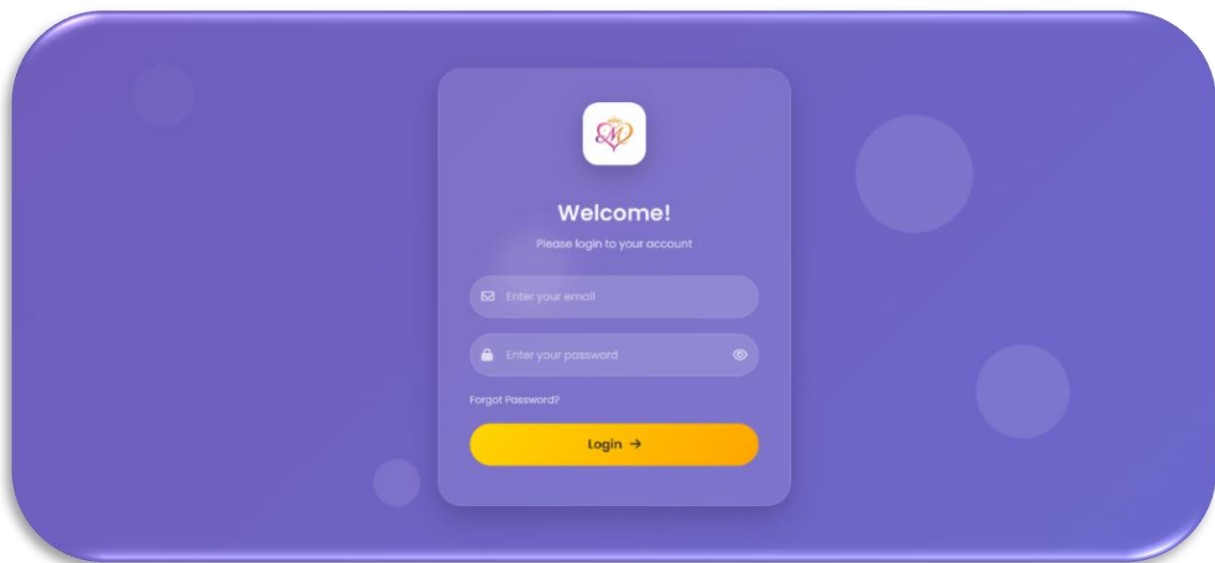
❖ User Login Page

Purpose:

Enables registered users to access the system securely.

Key Features:

- Email and password authentication
- Error handling for invalid credentials



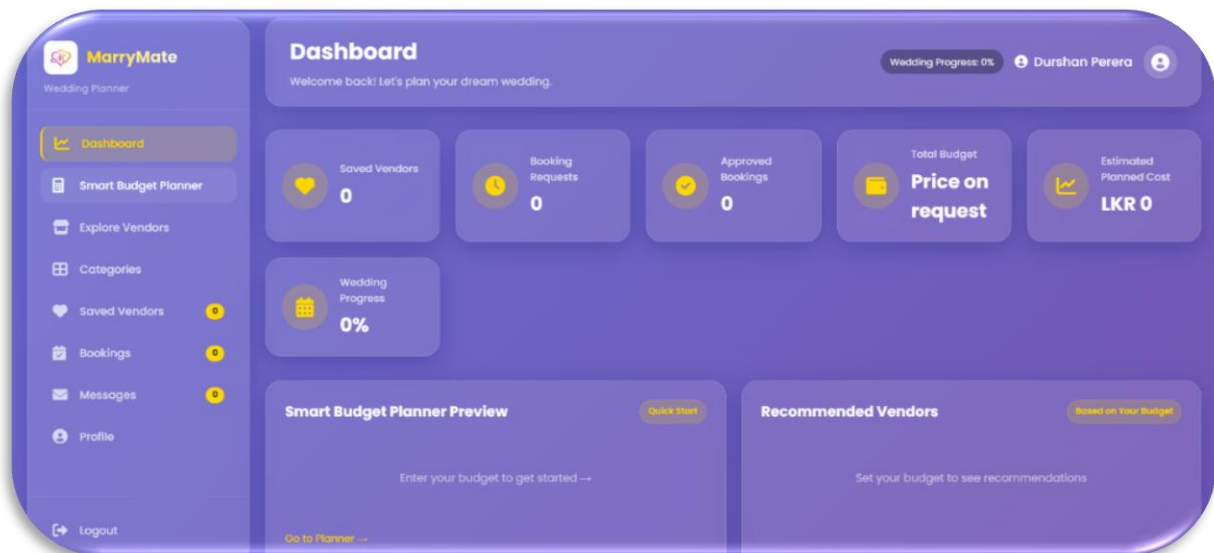
❖ Customer Dashboard

Purpose:

Serves as the main control panel for users.

Key Features:

- View bookings
- Manage wedding planning tasks
- Access saved vendors
- Budget tracking overview



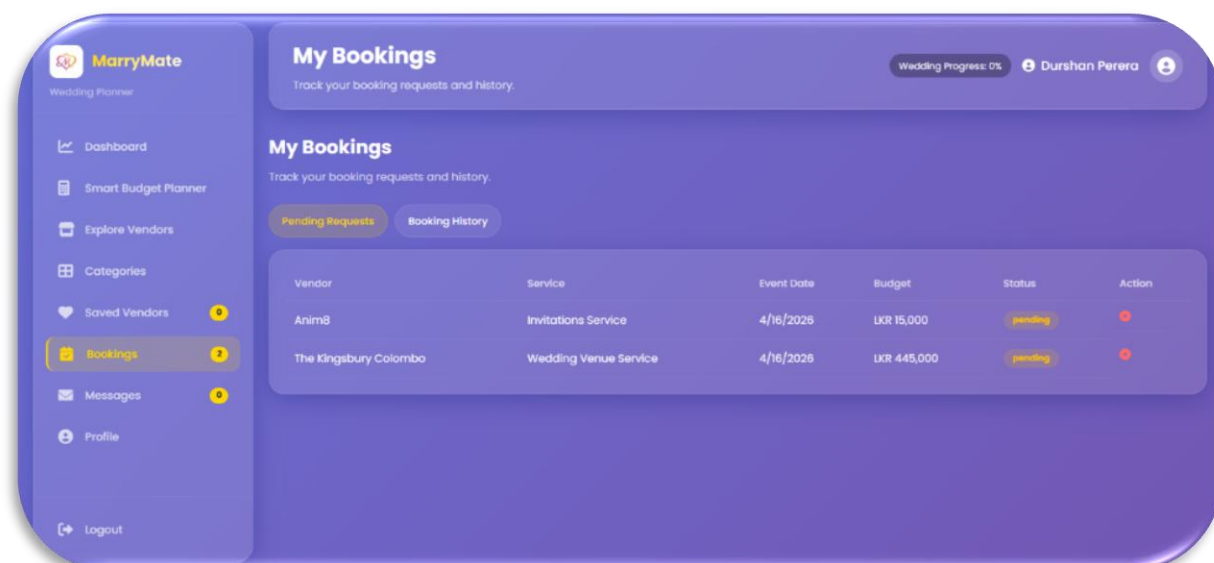
❖ Booking Management Page

Purpose:

Allows users to schedule and manage service bookings.

Key Features:

- Booking confirmation
- Booking history



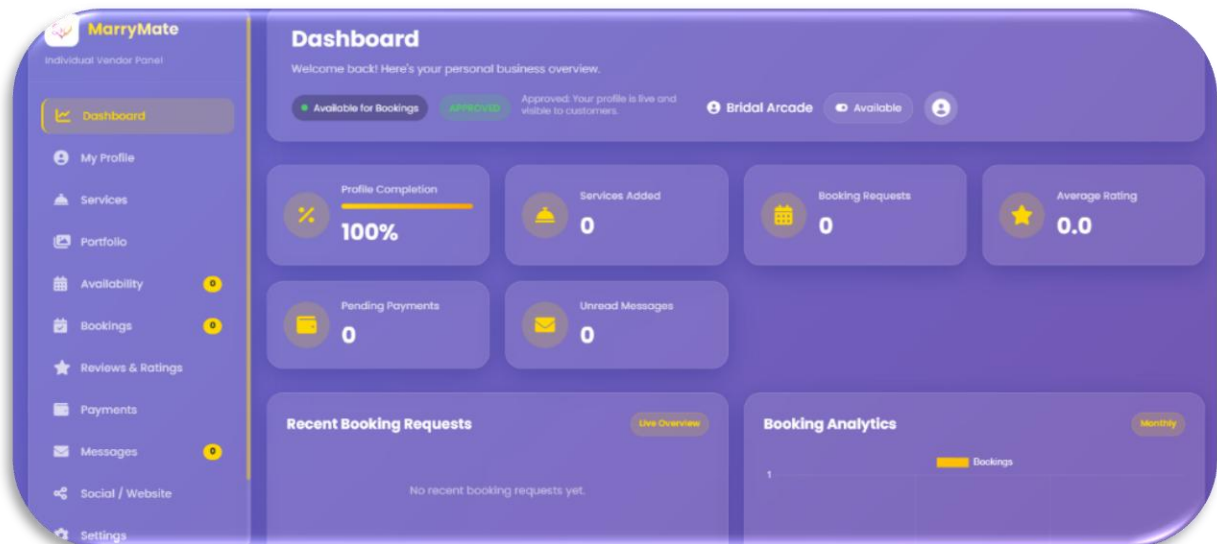
❖ Vendor Dashboard

Purpose:

Allows service providers to manage their services and bookings.

Key Features:

- Manage services and pricing
- Update availability
- View booking requests
- Monitor customer feedback



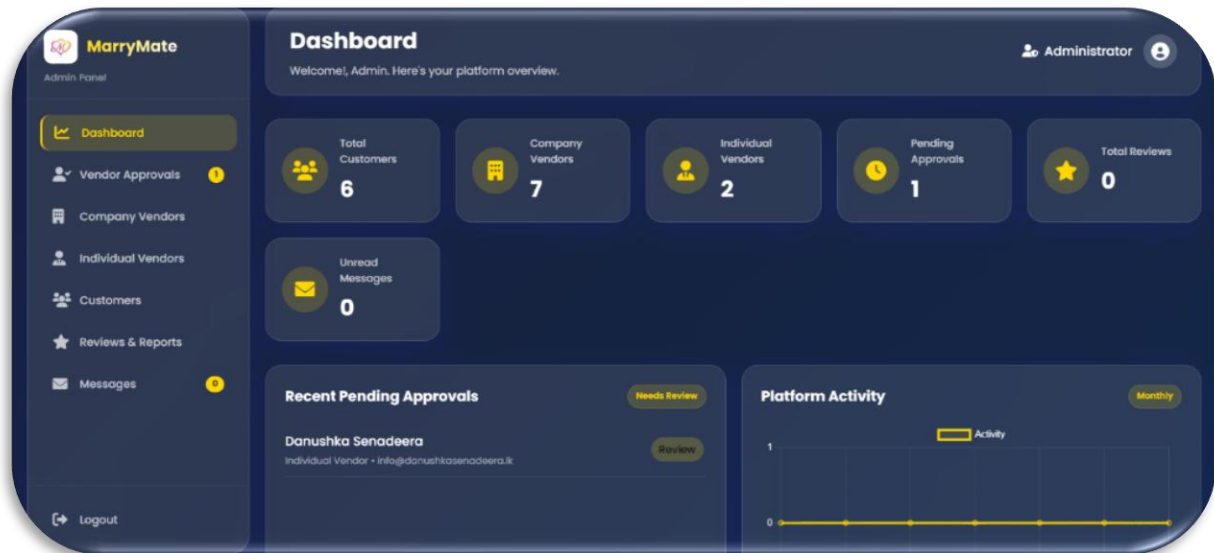
❖ Admin Dashboard

Purpose:

Provides administrative control over the platform.

Key Features:

- Vendor verification
- User management
- Category management
- System monitoring



4.3 Security and Access Control

Security is essential to protect user data and ensure system reliability.

Security is implemented using **Firestore Authentication**, which provides secure login and registration features. Role-based access control ensures that only authorized users can access certain functionalities:

- **Customers:** Can browse vendors, manage bookings, and track budgets.
- **Vendors:** Can manage services, respond to bookings, and view customer inquiries.
- **Admin:** Oversees system operations, approves vendors, and monitors activity logs.

4.4 Chapter Summary

Chapter 4 outlines the design and development of the MarryMate system, focusing on its architecture, interface, and security. The system is built on a client-server model, with a responsive frontend using HTML, CSS, and JavaScript, and a Firestore-powered backend that manages authentication, data storage, and real-time updates.

The interface design emphasizes simplicity and usability, offering tailored pages such as the home page, registration and login, customer and vendor dashboards, booking management, and an admin dashboard, each serving specific user needs. Security is ensured through Firestore Authentication and role-based access control, granting customers, vendors, and administrators appropriate permissions to perform their respective tasks. Overall, the chapter highlights a well-structured architecture, intuitive interface design, and robust security framework that together ensure the system's reliability, efficiency, and user-friendliness.

Chapter 5: Cost of Solution Development, Target Market &

Implementation Plan

5.1 Development Cost Estimation

The MarryMate system was developed using open-source technologies, which significantly reduced software costs. However, the development process involves costs related to hardware usage, internet access, electricity, and development effort.

Estimated Development Costs

Item	Description	Estimated Cost (LKR)
Hardware Usage	Laptop/PC usage and maintenance	15,000
Internet & Electricity	Connectivity and power consumption	8,000
Software Tools	Open-source tools (no licensing cost)	0
Development Effort	Time and labor contribution	25,000
Testing & Miscellaneous	Printing, documentation, testing	5,000

Total Estimated Cost: 53,000 LKR

Note: The development effort cost represents estimated labor value rather than actual payment.

5.2 Hardware and Software Requirements

❖ Hardware Requirements

For Development & Administration,

- Computer with minimum Intel i3 processor or equivalent
- 4 GB RAM (8 GB recommended)
- 500 GB storage
- Stable internet connection

For End Users,

- Smartphone, tablet, or computer
- Internet connectivity
- Modern web browser

❖ **Software Requirements**

Development Environment,

- Visual Studio Code
- Web browser (Chrome, Edge, Firefox)

Technologies Used,

- HTML, CSS, JavaScript
- Firebase

5.3 Target Market and Users

MarryMate is designed to serve a wide range of users involved in wedding planning.

Primary Users:

- Couples planning weddings
- Families organizing wedding events

Secondary Users:

- Wedding service providers
- Event planners
- Wedding coordinators

Service Providers Include:

- Wedding venues
- Photographers & videographers
- Caterers
- Decoration services
- Bridal dressing & salons
- Entertainment providers

❖ Market Opportunity

The wedding industry in Sri Lanka is a rapidly growing sector with increasing demand for convenience and digital services. Many service providers rely on social media promotion and manual booking methods, indicating a strong opportunity for a centralized digital marketplace.

MarryMate can support urban and suburban users by improving access to verified vendors and simplifying event planning.

5.4 Deployment

The MarryMate Wedding Planning Platform is deployed using **Firebase** Hosting, which provides a secure and reliable environment for web applications. Hosting on Firebase ensures that the system is easily accessible to users via any standard web browser, eliminating the need for local installations or complex setup procedures.

Firebase Hosting also supports secure **HTTPS** connections, which protect user data during transmission and ensure safe interactions between the system and its users. Furthermore, the platform leverages real-time updates, allowing changes in data such as vendor availability, bookings, or messages to be reflected instantly across all users' interfaces. This ensures a seamless and dynamic user experience, maintaining synchronization between customers, vendors, and administrators.

Overall, deploying the system on Firebase provides scalability, security, and accessibility, making it well-suited for modern web-based applications and supporting future enhancements without additional infrastructure overhead.

5.5 Chapter Summary

This chapter examined the practical feasibility of the MarryMate system by outlining the estimated development costs and highlighting the affordability achieved through the use of open-source technologies. It also identified the hardware and software requirements necessary for system development, deployment, and user access.

The target market analysis emphasized the relevance of the platform to couples, families, and wedding service providers, while recognizing the growing demand for convenient digital solutions within the wedding industry. In addition, an implementation and deployment strategy was presented to ensure smooth system rollout, user adoption, and ongoing maintenance.

These considerations demonstrate that MarryMate is a practical, cost-effective, and scalable solution suitable for real-world implementation.

Chapter 6: Discussion and Conclusion

6.1 Project Outcomes and Achievements

The development of MarryMate successfully demonstrated how a digital platform can simplify and improve the wedding planning process. The system integrates vendor discovery, service comparison, booking management, and communication into a single centralized platform.

Key achievements of the project include:

- Successfully developed a fully functional web-based wedding planning system.
- Implemented automated vendor data collection using web scraping.
- Designed a user-friendly interface with dashboards, booking pages, and vendor listings.
- Developed role-based access control to secure data and features.
- Creating separate dashboards for customers, vendors, and administrators

The system improves convenience and reduces the complexity associated with traditional wedding planning.

6.2 Challenges Faced and Solutions

- **Integration of Firebase Services and Handling Real-Time Update**

The system integrates multiple Firebase services including Authentication, Firestore Database, and Storage to provide a robust backend solution. Firebase enables real-time synchronization, ensuring that changes such as vendor bookings, messages, or budget updates are instantly reflected across all users' interfaces. This eliminates delays, reduces inconsistencies, and allows for smooth interactions between customers, vendors, and administrators. By leveraging Firebase's scalable infrastructure, the platform maintains high performance even as user activity increases.

- **Designing Responsive and Intuitive UI Across Devices**

A critical aspect of the system is its responsive and user-friendly interface. The platform is designed to function seamlessly on desktops, tablets, and mobile devices, adapting layouts and interactive elements to different screen sizes. Clear navigation, visually appealing dashboards, and accessible forms ensure that users can easily perform tasks without confusion. The use of modern design principles, such as glassmorphism and consistent color schemes, enhances usability and provides a professional aesthetic.

- **Debugging and Ensuring Seamless Interaction Between Modules**

During development, extensive debugging was conducted to ensure all system modules interact correctly. This includes verifying that authentication works with the database, vendor bookings are correctly processed, and notifications are delivered in real time. Rigorous testing helped identify and resolve potential conflicts between components, ensuring a smooth and error-free experience for all users. Integration issues were systematically addressed to maintain the stability and reliability of the platform.

6.3 Lessons Learned

- **Team Collaboration and Effective Communication**

Successful development of the MarryMate Wedding Planning Platform relied heavily on team collaboration and clear communication. Each team member was assigned specific responsibilities, such as frontend development, backend integration, database management, and UI/UX design. Regular meetings and progress updates ensured that all members were aligned on project goals, milestones, and deadlines. Effective communication helped in resolving challenges quickly, sharing ideas, and maintaining consistency across different modules, which was crucial for delivering a cohesive and fully functional system.

- **Real-World Application of Web Development, Database Management, and System Design**

This project provided an opportunity to apply real-world software engineering principles. Web development skills were utilized to create an interactive and responsive user interface using HTML, CSS, and JavaScript. Database management skills were applied in structuring and handling data using Firebase Firestore, ensuring data integrity, security, and real-time updates. System design concepts, including client-server architecture and modular design, were implemented to organize the platform efficiently. Through these applications, theoretical knowledge was transformed into practical skills that can be leveraged in professional software development environments.

- **Agile Methodology Improves Efficiency and Adaptability**

The project followed the Agile software development methodology, which emphasizes iterative development, continuous feedback, and flexibility. Agile allowed the team to break the project into smaller, manageable tasks and complete them in incremental sprints. This approach improved efficiency by enabling the team to focus on priority

features, test components in cycles, and make adjustments based on feedback. The iterative nature of Agile also ensured adaptability, allowing the project to evolve smoothly in response to emerging requirements or challenges.

6.4 Future Enhancements

- **Integration of AI-Based Vendor Recommendations for Personalized Suggestions**

One potential future enhancement for the MarryMate platform is the integration of Artificial Intelligence (AI) algorithms to provide personalized vendor recommendations. By analyzing user preferences, past interactions, budget constraints, and event requirements, the system can suggest vendors that are most relevant to each customer. This feature would improve decision-making, reduce time spent searching for suitable vendors, and enhance overall user satisfaction. AI-based recommendations also enable the system to learn from user behavior, continuously improving the accuracy and relevance of suggestions over time.

- **Development of a Mobile Application for On-the-Go Planning**

Another significant enhancement is the creation of a mobile application, allowing users to manage their wedding plans anytime and anywhere. A mobile app would provide on-the-go access to vendor listings, booking statuses, messages, and budget tracking, making the platform more convenient and accessible. By leveraging responsive design principles and mobile-friendly interfaces, the app ensures a seamless user experience, keeping customers, vendors, and event planners connected even when away from desktop devices.

- **Online Payment Gateway Integration for Smoother Transactions**

Integrating an online payment gateway is another key future improvement to streamline financial transactions within the platform. Customers would be able to securely pay vendors directly through the system, eliminating the need for external payment arrangements and reducing potential errors or delays. This integration would enhance trust, improve efficiency in booking confirmations, and create a seamless end-to-end wedding planning experience. Additionally, transaction records can be automatically stored for accounting and reporting purposes, further adding value for both users and vendors.

6.5 Conclusion

The MarryMate system was developed to address the challenges associated with traditional wedding planning by providing a centralized and user-friendly digital platform. Conventional planning methods often require couples to coordinate multiple vendors, manage schedules manually, and handle budgeting without clear visibility, which can lead to stress, miscommunication, and financial uncertainty. MarryMate was designed to simplify these complexities by integrating vendor discovery, service comparison, booking management, and communication into a single accessible platform.

Through the implementation of this system, users are able to explore verified service providers, compare services based on pricing and reviews, schedule appointments, and manage their wedding plans more efficiently. Vendors benefit from improved visibility and structured booking management, while administrators maintain system reliability through monitoring and verification processes. The adoption of client-server architecture and secure authentication mechanisms ensures that the platform is organized, scalable, and secure.

This project demonstrates how web-based technologies can be effectively applied to solve real-world coordination challenges within the wedding industry. Beyond technical implementation, the development process strengthened problem-solving abilities, teamwork, and project management skills. The resulting system provides a practical, cost-effective solution with the potential for future enhancements and real-world deployment. Overall, MarryMate contributes to modernizing wedding planning by transforming a traditionally complex process into a more efficient, transparent, and stress-free experience.

6.6 Chapter Summary

This chapter evaluated the overall outcomes of the MarryMate project and discussed how the system successfully addresses the challenges associated with traditional wedding planning. The achievements of the project were outlined, including the development of a centralized platform that improves vendor discovery, booking management, and communication. Challenges encountered during development were examined alongside the solutions implemented to overcome them.

The chapter also highlighted the technical and professional knowledge gained throughout the project, including web development skills, web scraping, teamwork, and problem-solving. Potential future enhancements were identified to expand system functionality and improve user experience. In summary, the MarryMate system demonstrates the effective use of web technologies to deliver a practical and scalable solution for modern wedding planning needs.

Chapter 7: References

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Chapter 8: Appendices